

494 Capstone Final Report Specifications

Format: This report should be completed using proper grammar and punctuation and be typo-free. APA format should be employed where applicable (e.g., figure labels, references).

1. Executive Summary

The Executive Summary is written typically for managers who need/want a broad understanding of the project and do not need/want a detailed understanding of the project. It should provide readers with an overview of the report before focusing on the details. This section should contain:

- A problem statement
- The major results and findings
- Your summary conclusions and recommendations

2. Introduction

Provide the “back story” on the project and the organization. Introduce the organization (the nature of its business) and discuss how this project was identified and why this project was chosen.

3. Statement of Engineering Problem

Discuss the project objectives and state the problem you examined. Explain why the issue presents a problem/challenge that needs to be resolved. Include any relevant supporting baseline data, as available, in the form of graphics, photos, and data summaries that demonstrate that the project was a problem deserving of attention.

4. Engineering Standards, Codes and Constraints

Identify all relevant standards, regulations and codes that apply directly (and indirectly) to the scope of your project and explain how these impacted or influenced your work. This should include any identified engineering standards, management standards, local, state, or federal codes as applicable. (Use bullet-point full sentences form.)

Describe key constraints that influenced the nature of your solution approach. (Use bullet-point full sentences form.)

5. Development of the Engineering Approach

Explain how you went about identifying, understanding and solving the problem you worked on: describe how the direction of the solution approach evolved chronologically from the project outset to conclusion, in terms of the below

- Key aspects of your understanding of the problem and/or constraints that emerged over time.
- How the solution approach was revised/adapted in response, including key decisions that shaped the path of the project.

Cite the Engineering Methods that were ultimately employed and explain the rationale for those choices. Use bullet point paragraph format, with each item containing the following information:

- *Engineering Method 1.* Explanation of rationale for method selection. Citation of the specific Department of Industrial and Systems Engineering course(s) where you gained the knowledge and skills required to successfully implement this method.

6. Activities and Analysis

This forms the core of the report. Provide a complete discussion of the activities and results for this project. Include the types of information/data collected, method of collection, results of collection (visuals where appropriate), analysis of results, and conclusions. Discussion should reflect a logical, evidence-based narrative of the project path and outcome.

7. Actions Taken/Recommended

Summarize the actions that have been taken, or that you have recommended be taken, to implement your findings. Identify criteria/metrics by which the results of your recommendation can be measured to demonstrate the value-added from your solution. Suggest methods for collecting that data.

Where relevant, evaluate what could go wrong with the solution and propose alternative viable solutions if needed.

APPENDIX: SELF-ASSESSMENT

In bullet point full-sentence format, describe:

- a) The professional traits most valued by employers (as per NACE or other sources from class discussion) that you selected as areas of focus for you at the start of the semester.
- b) The professional traits that you would now say that you did clearly demonstrate during your internship.
- c) Tell the story: input here your material from the “Professional Self-Assessment” class session.

Create a 90-second (ish) informal video under the title of “If Not for This Internship...”. Describe key lessons learned that that you believe significantly improved your understanding of the engineering workplace, and/or your own professional value. Do NOT identify the organization at which you interned.



Please indicate audiences, beyond the course instructor, with whom you would be willing for the instructor to share this video

- ABET accreditors
- Current and past ISE Dept Chair
- ISE Undergraduate Director
- ISE faculty
- ISE students